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Real-Time Air Canvas Drawing System Using OpenCV

A report on
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Submitted By
Keshav Shivhare -230962302
Nilesh Kumar -230962316

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
MANIPAL INSTITUTE OF TECHNOLOGY,
MANIPAL ACADEMY OF HIGHER EDUCATION
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Real-Time Air Canvas Drawing System Using OpenCV

Keshav Shivhare , Nilesh Kumar

[1] Department of Computer Science and Engineering

Manipal Institute of Technology Manipal Academy of Higher Education

Udupi, Karnataka, India

shivharekeshav26114@gmail.com , nileshmsd123@gmail.com

Abstract— This paper presents a comprehensive real-time air canvas drawing system enabling contactless digital artwork creation through hand gesture tracking using colored markers and webcam. The system integrates OpenCV-based HSV color detection, morphological filtering for noise reduction, contour analysis for accurate centroid localization, and efficient canvas state management supporting up to 20 undo/redo operations. Core features include four drawing modes (freehand, line, circle, rectangle), five-color palette, dual display backgrounds, and real-time geometric shape preview. Performance evaluation demonstrates 96.2% marker detection accuracy, consistent 30+ FPS real-time operation with sub-55ms end-to-end latency, comprehensive feature validation, and positive user satisfaction ratings. The modular software architecture facilitates future extensions including multi-user support, depth sensor integration, and advanced gesture recognition capabilities

Keywords— Air Canvas, Computer Vision, OpenCV, HSV Color Space, Hand Tracking, Gesture Recognition, Morphological Operations, Contour Detection, Real-time Drawing, Canvas State Management, Interactive Systems

I. INTRODUCTION

Contactless interaction technologies have become critically important in health-conscious environments, interactive educational settings, and professional presentation scenarios. Traditional digital drawing interfaces mandate physical contact with input devices such as mice, styluses, or touchpads, limiting applicability in sterile medical environments, shared classroom spaces, and public presentation venues. The COVID-19 pandemic further emphasized the necessity for touchless interaction systems that minimize surface contact transmission vectors while maintaining intuitive user experience [1]. Our Real-Time Air Canvas Drawing System addresses these critical needs and existing implementation limitations through: (1) robust HSV-based color detection with morphological filtering achieving 96.2% marker detection accuracy, (2) comprehensive drawing functionalities including geometric shape support with real-time preview capability, (3) efficient canvas state management enabling creative workflows through responsive undo/redo functionality, (4) dual display modes supporting diverse application scenarios, and (5) modular architecture design facilitating future enhancements. The system operates at 30+ FPS with sub-55ms latency, making it suitable for real-time interactive applications across educational, professional, medical, and creative domains.

II. LITERATURE REVIEW

A. Gesture Recognition and Hand Tracking Systems

Gesture recognition has undergone substantial evolution from hardware-based glove systems to camera-based approaches, signifi cantly enhancing user comfort and accessibility [1]. Contemporary implementations employ diverse sensing modalities including RGB cameras, Time-of-Flight depth sensors, and thermal imaging devices, utilizing techniques encompassing skin color detec tion, motion pattern analysis, hand skeleton estimation, and deep learning-based pose recognition models [10]. State-of-the-art systems achieve accuracy exceeding 90% for hand pose estimation when employing well-designed feature extractors combined with robust classification algorithms. Real-time tracking benefits substantially from contour-based methodologies and convex hull analysis, enhancing precision for fine finger movements in interactive drawing applications.

B. HSVColor Space Applications in Computer Vision

HSV (Hue, Saturation, Value) color space provides superior performance compared to RGB for object detection and tracking under varying illumination conditions [2]. Research investigations demonstrate HSV delivers substantially more precise color 1 segmentation for complex backgrounds, exhibiting close correspondence with manual segmentation results. The cylindrical co ordinate representation inherently separates chromatic content (hue and saturation) from achromatic intensity (value), enabling robust color-based

segmentation unachievable in the coupled RGB representation [3]. Comparative studies examining diverse color spaces reveal HSV's superior shadow suppression capabilities and enhanced tracking accuracy across diverse environmental conditions, making it ideal for marker detection systems.

C. Morphological Operations in Image Processing

Mathematical morphology constitutes fundamental techniques in image processing employed for noise reduction, shape analysis, and contour enhancement [4]. Elementary operations including dilation (expanding object regions) and erosion (shrinking object regions) establish the foundation for complex processes such as opening (erosion followed by dilation) and closing (dilation followed by erosion), effectively eliminating artifacts while preserving essential structural features [5]. Modern morphological processors optimized for real-time video processing achieve noise reduction rates approaching 97% through careful structuring element selection and operation sequencing. These operations prove essential for interactive computer vision applications where environmental noise, lighting variations, and sensor artifacts require systematic mitigation.

D. Contour Detection and Shape Analysis

Contour detection methodologies identify object boundaries within digital images enabling accurate spatial tracking, particularly effective when combined with preprocessing and morphological operations [6]. Standard algorithmic approaches involve binary mask generation through thresholding, connected region detection, and geometric shape analysis. Advanced techniques employ convex hull computation and geometric moment analysis to determine precise centroid coordinates, improving position track ing accuracy even in challenging environmental conditions. Area-based filtering combined with shape ratio validation further enhances marker recognition accuracy by eliminating false positive detections.

E. Air Canvas and Interactive Drawing Systems

Recent air canvas research emphasizes user-friendly interface design supporting natural creative expression with high gesture recognition accuracy. Earlier implementations utilized infrared LED tracking combined with Haar Cascade classifiers, demon strating effective real-time stylus tracking for interactive applications [7]. Contemporary systems integrate MediaPipe hand land mark detection and OpenCV image processing for millisecond-latency tracking supporting complex gesture vocabularies [8]. These systems demonstrate feasibility of marker-less hand tracking, though colored marker approaches often provide more robust performance under suboptimal lighting conditions and complex backgrounds.

F. Interactive System Design and User Interface Principles

User interface design for interactive systems requires careful consideration of control mapping, response latency, feedback mech anisms, and user mental models [9]. Dual display modes and comprehensive drawing tools extend system usability across diverse application domains from educational demonstrations to professional presentations. Efficient state management through opti mized data structures enables responsive undo/redo functionality critical for creative workflows. Research demonstrates that sys tems with sub-100ms latency provide satisfactory interactive experience for drawing applications, while those exceeding 200ms latency cause noticeable frustration.

III. OBJECTIVES

Objective 1

Develop a robust contactless drawing interface with adaptive color calibration capabilities, achieving 95%+ marker detection accuracy across diverse lighting conditions through HSV-based color detection combined with morphological processing, sup porting automatic environmental adjustment, providing user-friendly calibration interface, and maintaining consistent real-time performance exceeding 30 FPS.

Objective 2

Implement comprehensive drawing functionalities with dual-mode display and efficient state management, incorporating multiple drawing modes (freehand, line, circle, rectangle), enabling multi-level undo/redo history (up to 20 states), seamless mode toggling between camera overlay and whiteboard backgrounds, canvas persistence with automatic timestamping, and intuitive keyboard based control interface.

IV. METHODOLOGY

A. System Architecture Overview

The air canvas system comprises five interconnected functional modules processed sequentially on each captured video frame: (1) Video Capture and Preprocessing, (2) HSV-based Color Detection with Adaptive Thresholding, (3) Morphological Filtering Using Opening and Closing Operations, (4) Contour Detection and Centroid Localization via Geometric Moments, and (5) Drawing Engine with Comprehensive State Management. This pipeline architecture ensures computationally efficient processing while maintaining high marker detection accuracy and real-time responsiveness. The modular design allows independent optimization of each processing stage.*Title and Author Details*

B. Hardware and Software Specifications

Table 1: SYSTEM CONFIGURATION AND REQUIREMENTS

Component	**Specification**
Webcam	Minimum 640×480 pixels, 30 FPS capability
Processor	Intel Core i5 or equivalent (1.6 GHz+)
Memory	8GB RAM minimum for smooth operation
Storage	Sufficient for video capture and canvas history
Software	**Version Requirement**
Python	3.8 or higher
OpenCV	4.5+ with full module support
NumPy	Latest stable version
Operating System	Windows/Linux/macOS compatible

C. Color Detection Methodology

HSV color space conversion from BGR format employs OpenCV's cv2.cvtColor() function with cv2.COLOR BGR2HSV f lag for illumination-invariant color transformation. Blue marker detection utilizes carefully calibrated HSV ranges determined through extensive experimentation across varied lighting conditions: lower bound [100, 50, 50] and upper bound [130, 255, 255]. These parameters provide robust color segmentation with minimal false positives while accommodating marker color variations due to perspective distortion and ambient lighting changes. The HSV thresholds are designed to minimize background noise rejection while maintaining sensitivity to target marker colors

D. Morphological Processing Pipeline

The morphological processing sequence systematically eliminates noise artifacts while preserving marker structural integrity: (1) Create 5×5 structuring element kernel, (2) Apply morphological opening (erosion followed by dilation) for small noise removal, (3) Apply morphological closing (dilation followed by erosion) for internal hole filling. The two-stage approach maintains marker boundary definition while achieving approximately 97% noise reduction critical for accurate centroid localization. Opening operation removes small noise points by erosion followed by restoration through dilation, while closing operation seals internal gaps through dilation followed by boundary refinement through erosion.

E. Marker Detection and Centroid Calculation

Contour detection identifies connected regions within refined binary masks using OpenCV's findContours() function with RETR EXTERNAL retrieval mode for computational efficiency. The contour with largest area is selected as the primary marker candidate. Area-based filtering with 300-pixel minimum threshold eliminates spurious detections from background noise sources. Geometric moments enable precise centroid determination through: cx = M10/M00 and cy = M01/M00, where Mij represents spatial moments. This moment-based approach provides sub-pixel accuracy for centroid localization critical for smooth drawing trajectory rendering.

V. Experimental Setup

A. Marker Detection Algorithm Implementation

```
bluedef detect_finger(frame):
gray    gray#gray grayConvertgray grayBGRgray graytogray grayHSVgray grayforgray
        grayilluminationgray grayinvariance
    hsv = cv2.cvtColor(frame, cv2.COLOR_BGR2HSV)

gray    gray#gray grayApplygray grayHSVgray graythresholdinggray grayforgray graymarkergray
        grayisolation
    mask = cv2.inRange(hsv, lower_hsv, upper_hsv)

gray    gray#gray grayCreategray gray5grayx5gray graykernelgray grayforgray graymorphological
        gray grayoperations
    kernel = np.ones((5,5), np.uint8)

gray    gray#gray grayApplygray grayopeninggray gray(grayerosiongray graythengray graydilation
        gray)
    mask = cv2.morphologyEx(mask,
                            cv2.MORPH_OPEN, kernel)

gray    gray#gray grayApplygray grayclosinggray gray(graydilationgray graythengray grayerosion
        gray)
    mask = cv2.morphologyEx(mask,
                            cv2.MORPH_CLOSE, kernel)

gray    gray#gray grayFindgray graycontoursgray grayingray grayprocessedgray graymask
    contours, _ = cv2.findContours(mask,
                      cv2.RETR_EXTERNAL,
                      cv2.CHAIN_APPROX_SIMPLE)

    blueif bluenot contours:
        blueturn None

gray    gray#gray graySelectgray graycontourgray graywithgray graymaximumgray grayarea
    c = bluemax(contours, key=cv2.contourArea)

gray    gray#gray grayFiltergray graybygray grayminimumgray grayareagray graythreshold
    blueif cv2.contourArea(c) < 300:
        blueturn None

gray    gray#gray grayCalculategray graygeometricgray graymoments
    M = cv2.moments(c)
    blueif M["m00"] == 0:
        blueturn None

gray    gray#gray grayComputegray graycentroidgray graycoordinates
    cx = blueint(M["m10"]/M["m00"])
    cy = blueint(M["m01"]/M["m00"])

    blueturn (cx, cy)
```

B. Drawing Engine and Canvas State Management

The drawing engine implements four distinct operational modes: (1) Normal Mode enabling freehand continuous drawing with adaptive line smoothing, (2) Line Mode supporting two-point straight line definition with real-time preview rendering, (3) Circle Mode implementing center-radius circle definition with continuous preview updates, and (4) Rectangle Mode supporting corner defined rectangular shapes with live preview functionality. Canvas states are maintained using efficient stack-based approach with bounded history depth

(maximum 20 states). Each drawing action saves current canvas state to undo history. Undo operations restore previous state while saving current state to redo history. New drawing actions automatically clear redo history, ensuring intuitive user interaction. This design maintains efficient memory usage through state depth limitation while enabling rapid state transitions.

C. User Interface Control Mapping

Table 2: COMPREHENSIVE KEYBOARD CONTROL MAPPINGS

Key	Function	Key	Function
P	Toggle pen on/off	X	Clear canvas
B	Toggle background	U	Undo action
N	Normal drawing	Y	Redo action
L	Line drawing	F	Fullscreen
O	Circle drawing	Q	Quit
R	Rectangle drawing	Esc	Exit fullscreen
1-5	Select color		

VI. RESULTS AND DISCUSSION

A. Performance Metrics Summary

Table 3: COMPREHENSIVE SYSTEM PERFORMANCE METRICS

Metric	Value	Unit
Average Frame Rate	31.4	FPS
Detection Latency	27.5	milliseconds
End-to-End Latency	50.0	milliseconds
Marker Detection Accuracy	96.2	percent
False Positive Rate	2.1	percent
Memory Consumption	200	MB
Undo/Redo Response	17.5	milliseconds
Mode Switching Latency	10.0	milliseconds
Canvas State Size	2.3	MB

B. Detection Accuracy Under Varying Environmental Conditions

Table 4: DETECTION ACCURACY ACROSS DIVERSE ENVIRONMENTAL CONDITIONS

Environmental Condition	Accuracy (%)	FPS	Reliability
Optimal Controlled Lighting	98.5	33	Excellent
Natural Daylight	96.8	32	Excellent
Low Illumination	91.2	31	Good
Complex Background	94.3	30	Good
Fast Movement	89.7	31	Acceptable
Average Performance	**94.1**	**31.4**	**Excellent**

The system maintains robust performance (94%+ average accuracy) across diverse environmental conditions, demonstrating ef fective real-world applicability. Optimal controlled lighting achieves maximum 98.5% accuracy while challenging low-light conditions maintain acceptable 91.2% performance, validating HSV color space robustness

C. Morphological Operations Impact Analysis

Table 5: MORPHOLOGICAL PROCESSING EFFECTIVENESS ANALYSIS

Configuration	Accuracy (%)	False Pos. (%)	Improvement
No Morphology	78.3	14.7	Baseline
Opening Only	88.6	6.8	+10.3%
Closing Only	84.2	9.3	+5.9%
Opening + Closing	96.2	2.1	+17.9%

Morphological operations demonstrate substantial impact on detection performance, achieving 17.9% accuracy improvement and 12.6% false positive reduction compared to unprocessed binary masks. The combination of opening (noise removal) and closing (hole filling) provides optimal performance validation.

D. Feature Functionality Validation Results

All implemented features successfully passed rigorous functionality validation testing: (1) Freehand drawing mode—smooth con tinuous line rendering without discontinuities, (2) Line drawing mode—straight line accuracy within 1 pixel, (3) Circle drawing mode—circular shape consistency within 2%, (4) Rectangle drawing mode—right angle preservation within 0.5 degrees, (5) Color selection—accurate rendering across five-color palette, (6) Undo functionality—correct state restoration across 20-state history, (7) Redo functionality—forward state restoration without data corruption, (8) Canvas clear—complete reset to blank state, (9) Pen toggle—responsive enable/disable switching, (10) Background toggle—seamless mode transition, (11) Fullscreen mode—proper display scaling without artifacts.

E. Comparative Analysis with Existing Systems.

Table 6: COMPARATIVE PERFORMANCE ANALYSIS WITH EXISTING IMPLEMENTATIONS

System	Method	Acc.	FPS	Shapes	Undo	Dual
Infrared LED	Haar Cascade	92	25	No	No	No
MediaPipe	Hand Landmarks	94	28	Limited	No	No
HSV Basic	Basic Tracking	89	30	No	No	No
Proposed	**HSV+Morphology**	**96.2**	**31**	**Yes**	**Yes**	**Yes**

The proposed system outperforms existing implementations in detection accuracy (96.2% vs 92-94%), feature completeness (4 shapes vs 0-1 shapes), and operational robustness across environmental conditions.

VII. CONCLUSIONS

This paper presented a comprehensive real-time air canvas drawing system successfully achieving 96.2% marker detection accu racy through HSV color detection combined with morphological filtering. The system maintains 30+ FPS real-time performance with sub-55ms end-to-end latency, implements comprehensive drawing functionality across four distinct modes with five-color palette selection, and provides efficient 20-state undo/redo history with sub-20ms response times. Experimental evaluation con f irms system robustness across diverse environmental conditions (94.1% average accuracy) and successful validation of all imple mented features. The modular software architecture design facilitates future enhancements toward more sophisticated contactless interaction systems suitable for educational demonstrations, professional presentations, healthcare applications, and creative ex pression domains

VIII. FUTUREWORK

Future research directions include: (1) Multi-user support through simultaneous multiple marker tracking with color differen tiation, (2) Depth sensor integration (Intel RealSense, Microsoft Kinect) enabling natural pen ”up/down” gesture detection, (3) Advanced gesture recognition expanding interaction vocabulary beyond drawing, (4) Adaptive color calibration system eliminat ing manual threshold adjustment requirements, (5) Temporal trajectory smoothing reducing hand tremor effects while maintaining responsiveness, (6) Mobile platform deployment optimizing for smartphone processors and camera specifications, (7) Machine learning-based marker detection improving robustness across diverse marker appearances and backgrounds, (8) Eraser function ality and layer management supporting complex digital artwork creation

IX. CONTRIBUTIONS

Contributions of Nilesh Kumar (230962316)

Nilesh Kumar significantly contributed to the system architecture and implementation, including designing and developing the morphological processing pipeline that uses opening and closing operations for noise reduction, achieving a 17.9 percent accu racy improvement. He implemented contour detection and centroid calculation algorithms using OpenCV moment analysis for precise sub-pixel accuracy, developed and rigorously tested the marker detection module securing 96.2 percent detection accuracy across diverse environments, and implemented dual display mode functionality for seamless transitions between camera overlay and whiteboard backgrounds. He conducted comprehensive experimental evaluations under varied lighting conditions,

performed statistical analysis of performance metrics, and documented the technical details, algorithmic approaches, and performance char acterization.

Contributions of Keshav Shivhare (230962302)

Keshav Shivhare focused on developing the drawing engine and user interface design, implementing four distinct drawing modes (freehand, line, circle, rectangle) with real-time preview, and creating a canvas state management system with efficient stack-based undo/redo functionality supporting 20 states and sub-20ms response times. He also developed an intuitive keyboard control system with over 14 commands, designed the color palette system for five drawing colors with visual feedback, developed the video capture and preprocessing module that includes frame stabilization and adaptive lighting adjustment, and led feature validation and comparative analysis with existing air canvas implementations

Joint Team Responsibilities

Both team members collaboratively: (1) Defined detailed project objectives and system requirements; (2) Conducted compre hensive literature review on gesture recognition, color spaces, and morphological operations; (3) Planned and executed system integration integrating individual component modules; (4) Performed extensive end-to-end testing and systematic debugging; (5) Prepared comprehensive project documentation and final technical report; (6) Participated in code review sessions identifying optimization opportunities; (7) Collaborated on performance evaluation and presentation preparation.

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